Exercise 3

1. Table : products

product_id	product_name	Price
1	Laptop	1200.00
2	Phone	800.00
3	Keyboard	45.00
4	Monitor	300.00
5	Mouse	25.00

'Expensive' if price > 1000
'Mid-range' if price between 100 and 1000
'Budget' if price < 100

```
SELECT    product_name,
          price
FROM      products
```

```
SELECT    product_name,
          price,
          CASE
            WHEN price > 1000 THEN 'Expensive'
            WHEN price BETWEEN 100 and 1000 THEN 'Mid-range'
            WHEN price < 100 THEN 'Budget'
          END AS price_category
FROM      products ;
```

product_name	price	price_category
Laptop	1200	Expensive
Phone	800	Mid-range
Keyboard	45	Budget
Monitor	300	Mid-range
Mouse	25	Budget

2. Table : Orders

order_id	customer_name	amount
1	Alice	150.00
2	Bob	560.00
3	Charlie	999.99
4	Diana	45.50
5	Ethan	1200.00

- 'High value' for orders ≥ 1000
- 'Medium value' for 500 - 999.99
- 'Low value' for orders < 500

```
SELECT  customer_name,
        amount,
        CASE
            WHEN  amount ≥ 1000  THEN 'High value'
            WHEN  amount BETWEEN 500 AND 999.99 THEN 'Malium Value'
            WHEN  amount < 500  THEN 'Low Value'
        END  AS  order_value_category
FROM  orders ;
```

Customer_name	amount	order_value_category
Alice	150.00	Low value
Bob	560.00	Medium value
Charlie	999.99	Medium value
Diana	45.50	Low value
Ethan	1200.00	High value

3. Table : employees

emp_id	emp_name	department	salary
1	John	IT	85000
2	Sara	HR	60000
3	Mark	IT	75000
4	Lucy	Finance	95000
5	Tom	HR	55000

- If in 'IT' and salary > 80000 → 'Senior IT'
- If in 'HR' and salary > 55000 → 'Experienced HR'
- Otherwise → 'Staff'

```
SELECT    emp_name,
          department,
          salary,
    CASE
        WHEN department = 'IT' AND salary > 80000 THEN 'Senior IT'
        WHEN department = 'HR' AND salary > 55000 THEN 'Experienced HR'
        ELSE 'Staff'
        END AS position_level
FROM      employees ;
```

emp_name	department	salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

4. Table : students

student_id	student_name	score
1	Anna	92
2	Ben	76
3	Cara	59
4	David	83
5	Ella	68

- ≥ 90 : 'A'
- 80 - 89 : 'B'
- 70 - 79 : 'C'
- 60 - 69 : 'D'
- < 60 : 'F'

```
SELECT    student_name,
          score,
     CASE
      WHEN   score ≥ 90 THEN 'A'
      WHEN   score BETWEEN 80 AND 90 THEN 'B'
      WHEN   score BETWEEN 70 AND 79 THEN 'C'
      WHEN   score BETWEEN 60 AND 69 THEN 'D'
      ELSE   'F'
      END AS   grade
FROM      students ;
```

Student_name	Score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

5. Table : deliveries

```
SELECT delivery_id,
       delivery_time_minutes,
       CASE
        WHEN delivery_time_minutes <= 30 THEN 'Fast'
        WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN 'On Time'
        WHEN delivery_time_minutes = 60 THEN 'Late'
        END AS performance
FROM deliveries;
```

delivery_id	delivery_time_minutes	performance
1	45	On Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

6. Table : tickets

```
SELECT issue_type,
       priority,
       CASE
       WHEN priority = 3 THEN 'High'
       WHEN priority = 2 THEN 'Medium'
       WHEN priority = 1 THEN 'Low'
       END AS priority_label
FROM tickets;
```

issue_type	priority	priority_level
Login issue	1	Low
Server down	3	~~Medium~~ High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

7. Table : attendance

```
SELECT    student_id,
          attendance_percentage
```

7. Table : attendance

```
SELECT    student_id,
          (days_present * 100 / total_days) AS attendance_percentage,
  CASE
  WHEN (days_present * 100 / total_days) >= 90 THEN 'Excellent'
  WHEN (days_present * 100 / total_days) BETWEEN 75 AND 89 THEN 'Good'
  ELSE 'Needs Improvement'
  END AS attendance_status
FROM      attendance;
```

student_id	attendance_percentage	attendance_status
1	90	Excellent
2	60	Needs Improvement
3	96	Excellent
4	50	Needs Improvement
5	100	Excellent

8. Table : products_inventory

```
SELECT  product_id,
        stock_qty,
        CASE
        WHEN  stock = 0  THEN  ~~sold~~ 'out of stock'
        WHEN  stock  BETWEEN 1 AND 5 THEN 'Low stock'
        ELSE  'in stock'
        END AS  stock_status
FROM    products_inventory;
```

product_id	stock_qty	stock_status
1	5	Low Stock
2	0	Out of Stock
3	25	In stock
4	10	In Stock
5	3	Low Stock

9. Table : classes

```
SELECT  subject,
        enrolled_students,
        CASE
        WHEN  enrolled_students >= 25  THEN  'Large'
        WHEN  enrolled_students BETWEEN 10 AND 24  THEN  'Medium'
        ELSE  'small'
        END AS  class_size_category
FROM    classes;
```

Subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

10. Table : payments

```
SELECT    payment_id,
          payment_method,
          amount,
      CASE
      WHEN    payment_method = 'cash' AND amount ≥ 200 THEN 'Eligible for Discount'
      ELSE    'Not Eligible'
      END AS    discount_eligibility
FROM      payments;
```

payment_id	payment_method	amount	discount_eligibility
1	Card	50.00	Not Eligible
2	Cash	200.00	Eligible for Discount
3	Card	150.00	Not Eligible
4	PayPal	75.00	Not Eligible
5	Cash	800.00	Eligible for Discount